

FREEPORT, TX, USA

WMO: 998189

Lat: 28.9357N

Lon: 95.2941W

Elev: 4

StdP: 101.28

Time Zone: -6.00 (NAC)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 0.7	(c) 3.0	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 10.3	(k) 7.9	(l) 8.9	(m) 12.3	(n) 4.6	(o) 20	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 3.0	(c) 31.2	(d) N/A	(e) 30.6	(f) N/A	(g) 30.3	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 4.3	(p) 190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 9.3	(b) 8.5	(c) 7.9	(e) -1.0	(f) 33.6	(g) 2.8	(h) 1.3	(i) -3.0	(j) 34.5	(k) -4.6	(l) 35.2	(m) -6.2	(n) 36.0	(o) -8.2	(p) 36.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	21.3	12.0	13.7	16.9	20.8	24.5	27.8	28.6	28.8	26.7	23.1	17.4	14.2		
	DBStd	6.78	4.50	4.49	3.80	2.86	2.32	1.27	1.01	1.16	1.94	3.48	4.36	4.43		
	HDD10.0	76	33	19	4	0	0	0	0	0	0	0	4	14		
	HDD18.3	625	198	134	69	12	1	0	0	0	0	7	67	138		
	CDD10.0	4184	94	122	219	324	448	535	578	583	501	406	228	146		
	CDD18.3	1692	1	4	26	86	191	285	319	325	251	154	40	10		
	CDH23.3	17332	0	0	5	167	1364	3383	4084	4264	2782	1233	49	1		
	CDH26.7	5703	0	0	1	15	83	1099	1656	1856	848	146	0	0		
(14) Wind	WSAvg	4.3	3.9	4.1	4.0	4.8	5.4	4.9	4.6	4.3	4.2	4.1	3.6	3.5		
(15) Precipitation	PrecAvg	1219	106	64	82	83	81	113	116	92	158	116	123	96		
	PrecMax	1771	327	232	235	209	176	295	319	240	425	340	405	213		
	PrecMin	589	4	8	15	5	0	10	6	19	27	1	20	23		
	PrecStd	296	74	56	57	61	53	74	80	63	111	92	93	61		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.1	22.1	23.8	28.8	29.1	32.9	32.1	32.4	31.3	29.6	25.6	23.0		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	19.7	20.9	22.8	25.5	28.0	30.5	30.4	31.2	30.2	28.9	24.7	22.0		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	18.8	19.8	22.2	24.8	27.5	29.9	30.2	30.8	29.9	28.3	24.0	21.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	DB	17.9	18.9	21.6	24.3	27.1	29.5	30.0	30.5	29.5	27.5	23.2	20.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.6	5.4	4.5	4.1	3.1	2.6	2.5	3.0	4.2	5.3	6.0	6.2		
		MCDBR	6.2	5.0	4.0	4.5	3.2	3.2	2.7	3.1	4.0	4.4	4.8	6.0		
	5% WB	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.353	0.367	0.381	0.437	0.478	0.461	0.486	0.470	0.450	0.387	0.379	0.361		
	taud		2.505	2.475	2.424	2.271	2.189	2.299	2.244	2.290	2.314	2.478	2.449	2.489		
	Ebn at Noon		885	902	908	863	824	834	811	823	829	868	847	857		
	Edh at Noon		91	102	114	137	149	132	140	132	125	100	96	89		
(44) All-Sky Solar Radiation	RadAvg		2.91	3.35	4.57	5.48	6.21	6.50	6.21	5.95	5.06	4.47	3.37	2.60		
	RadStd		0.30	0.54	0.41	0.44	0.41	0.45	0.57	0.42	0.51	0.48	0.34	0.31		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (28 neighbors)	+0.33	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+116	+100

Nomenclature: See separate page